Energy-Aware Adaptive IoT Sensing Using Edge Intelligence

IoT and Industrial 4.0 course project · Problem Statement 4

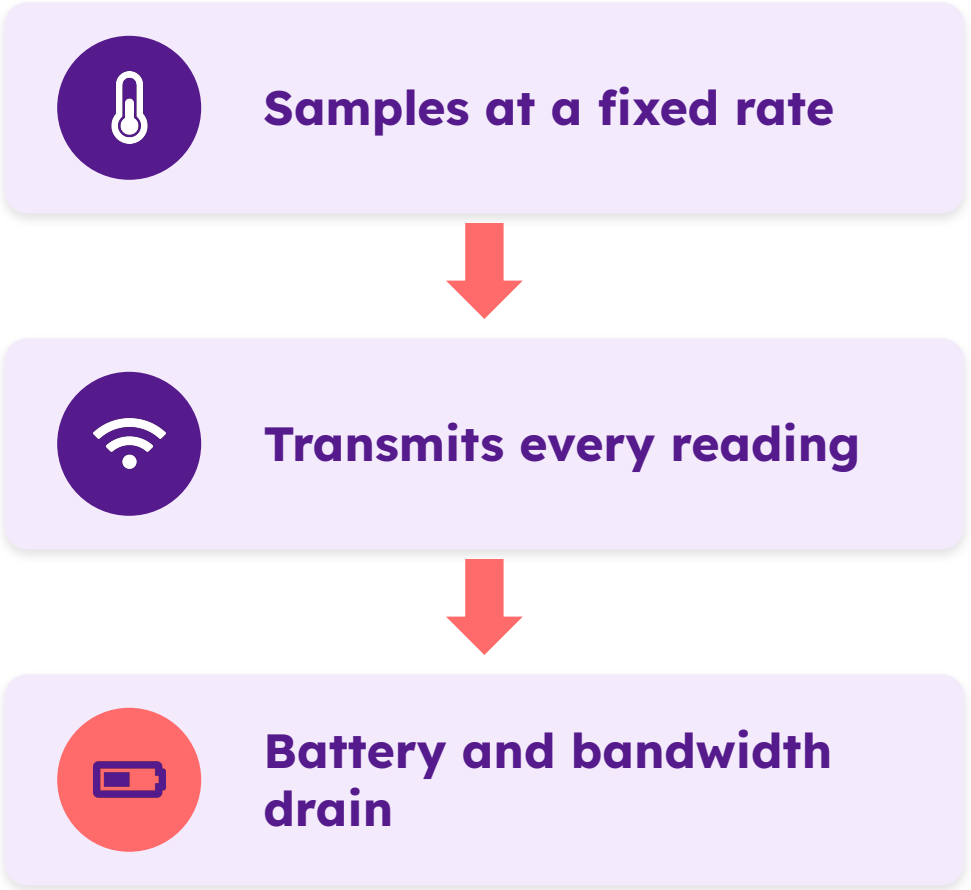
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Problem and motivation

Why continuous sensing is wasteful

- IoT nodes usually sample and transmit continuously
- This continues even when the monitored system is stable
- Result: wasted battery energy and network bandwidth
- Most transmitted data is redundant
- Goal: sense and send only as much as the condition demands



Research gap

What existing approaches leave unanswered

Fixed rate

Same sampling and radio use in calm and abnormal periods, so energy is spent even when nothing changes.

Simple thresholds

Switch rates when a limit is crossed, but ignore trends and can miss slow changes in the signal.

Cloud-only decisions

Add latency and network traffic for decisions that a small edge node can make locally.

Gap: few low-cost studies quantify the energy saved against detection accuracy and latency, using real data.

Research question and objectives

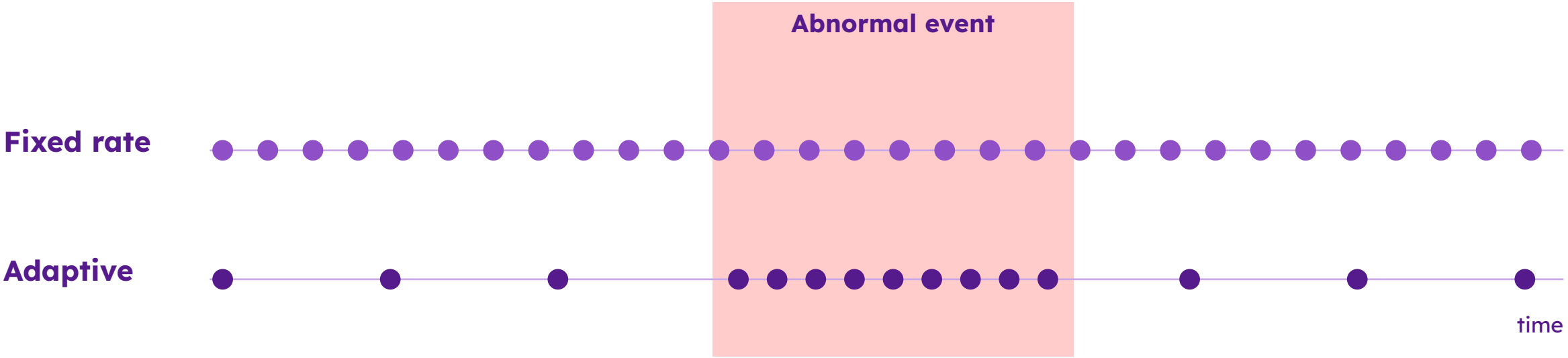
One question, four measurable objectives

How much energy and communication overhead can be reduced without significantly affecting detection accuracy?

1. Build an ESP32 node that adapts sampling, transmission and processing
2. Collect real data under normal and abnormal conditions
3. Compare fixed-rate, rule-based and adaptive approaches
4. Report energy, packets, bandwidth, accuracy, latency and battery life

Proposed approach

Sense more only when the environment changes



Normal condition

Low sampling rate and low communication.

Abnormal condition

High sampling rate and immediate transmission.

Edge decision

The ESP32 picks the mode locally with a lightweight algorithm.

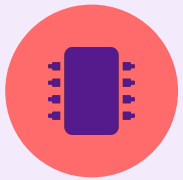
System architecture

From sensor reading to measured energy cost



1 Sense

DHT22 temperature and humidity plus MPU6050 motion



2 Edge

ESP32 compares readings and selects the mode



3 Send

Wi-Fi link carrying MQTT messages



4 Log

Laptop broker stores every record for the dataset



5 Measure

INA219 logs current to compute energy per mode



Feedback loop: the ESP32 chooses the next sampling interval from the latest readings, and INA219 measurements confirm the energy cost of each mode.

Hardware

Low-cost parts, chosen for the research question

ESP32

Edge node: runs the mode logic and the Wi-Fi link

DHT22

Slow environmental sensor, one reading per 2 s at best

MPU6050

Fast motion sensor that triggers abnormal events

INA219

Measures real current and energy of the node

18650 + TP4056

Battery runs and lifetime estimate from measured current

Breadboard kit

Solder-free prototyping with jumper wires

Optional: LEDs, push button for manual event marking, multimeter for cross-checking.

Adaptive sampling logic

A simple three-state controller running on the edge



Sample every 30 to 60 s, sleep between samples, send a periodic summary.

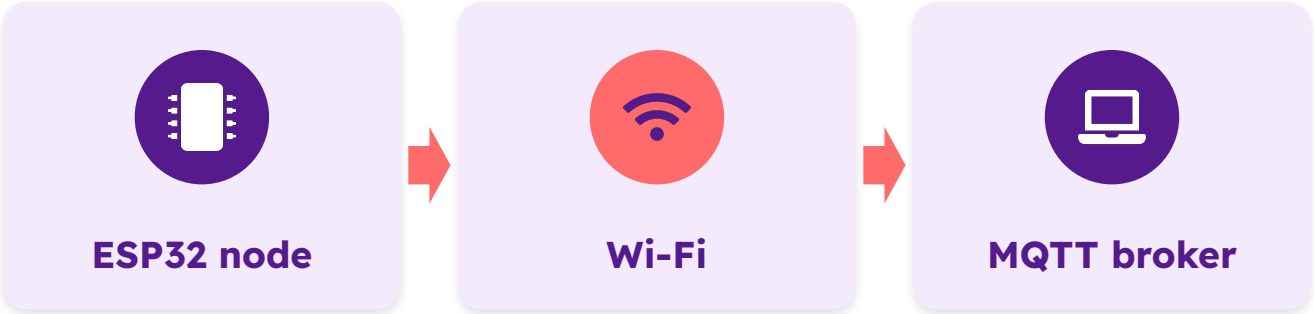
Rate of change or limit exceeded: sample every 2 s and send immediately.

After N stable readings, step back to the low rate.

Thresholds, intervals and N are tunable parameters chosen from experiments. DHT22 limits the fast rate to about one reading per 2 s.

Communication and edge processing

Sleep most of the time, transmit only when needed



Duty cycle: sleep, wake, send



- Wi-Fi with MQTT: lightweight publish/subscribe, broker on a laptop
- Light or deep sleep between samples to realise the savings
- Edge decides locally; sends on events plus a periodic heartbeat
- Trade-off: Wi-Fi reconnection costs energy, so batching matters

Dataset and experiments

Real data with labelled events, not simulated values

Timestamp	Temperature	Humidity	Motion	Current	Mode	Event label
yyyy-mm-dd hh:mm:ss	°C	%	g	mA	NORMAL / ALERT	0 / 1

Logged fields

Timestamp, temperature, humidity, motion, current, operating mode.

Test conditions

Stable room, heat source, breathing on the sensor, shaking the board.

Ground truth

Button press or manual log marks the start and end of each event.

Baselines

Compared fairly on the same hardware

Baseline 1

Fixed-rate sampling and transmission.

Baseline 2

Rule-based threshold switching.

Proposed

Adaptive edge approach using rate of change and limits.

All approaches run on the same ESP32, in separate runs, under the same conditions.

Evaluation metrics

How success will be measured

Energy

INA219 current multiplied by supply voltage and time, per mode

Traffic volume

MQTT message count and bytes sent, compared with fixed rate

Detection accuracy

Detected events against button-marked ground truth

Detection latency

Time from event start to the first detection

Battery lifetime

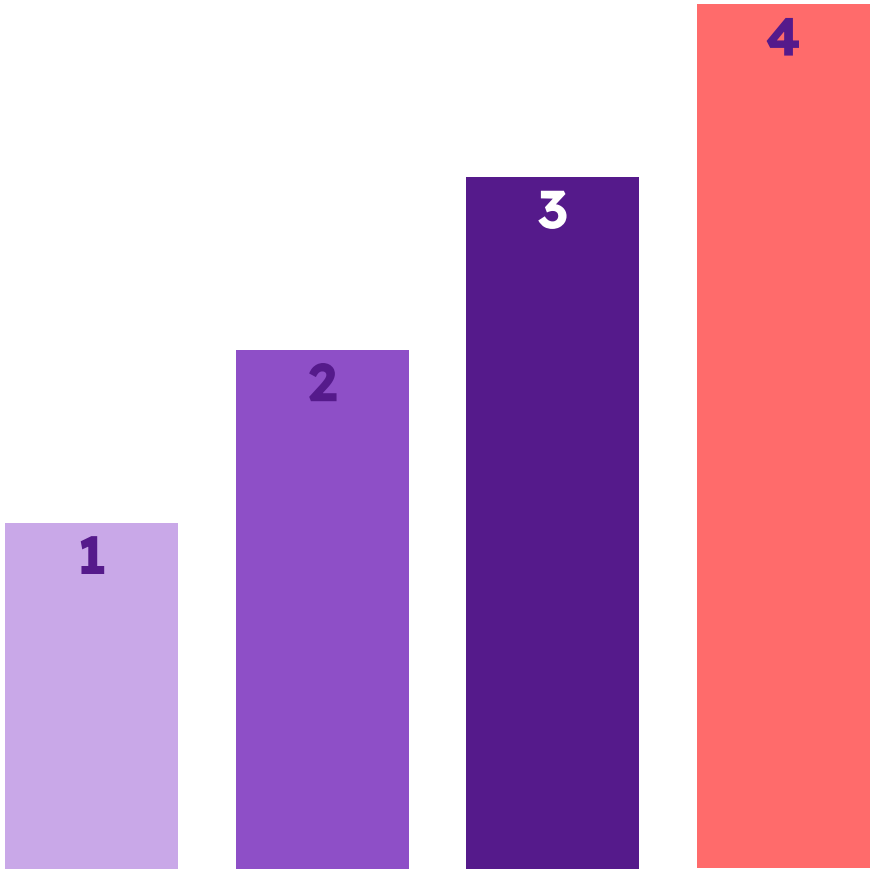
Assumed capacity divided by measured average current

Energy saving

Percentage reduction of the adaptive node against the fixed-rate baseline

Ablation study plan

Which component actually contributes to the saving?



Components added one at a time

Configuration	Energy	Accuracy	Latency
1 Fixed rate	[TBD]	[TBD]	[TBD]
2 Adaptive sampling only	[TBD]	[TBD]	[TBD]
3 + adaptive transmission	[TBD]	[TBD]	[TBD]
4 + adaptive processing (full)	[TBD]	[TBD]	[TBD]

Values are filled in after experiments.

Limitations and risks

Being open about where the method can fail

Detection delay

Low-rate mode can delay detection by up to one full sampling interval. This is the core trade-off the project measures.

Sensor speed

DHT22 is slow, so the MPU6050 widens the range between low and high sampling.

Wi-Fi cost

Reconnection overhead can offset sampling savings, so transmissions are batched.

Measurement

Without the INA219, energy would be estimated from datasheet values, not measured.

Expected contribution

What this project adds

1

An adaptive sensing and communication strategy for energy-efficient IoT

2

Quantified energy, bandwidth and accuracy trade-offs from real data

3

A low-cost, reproducible ESP32 implementation

Next: literature review → system design → prototype and dataset

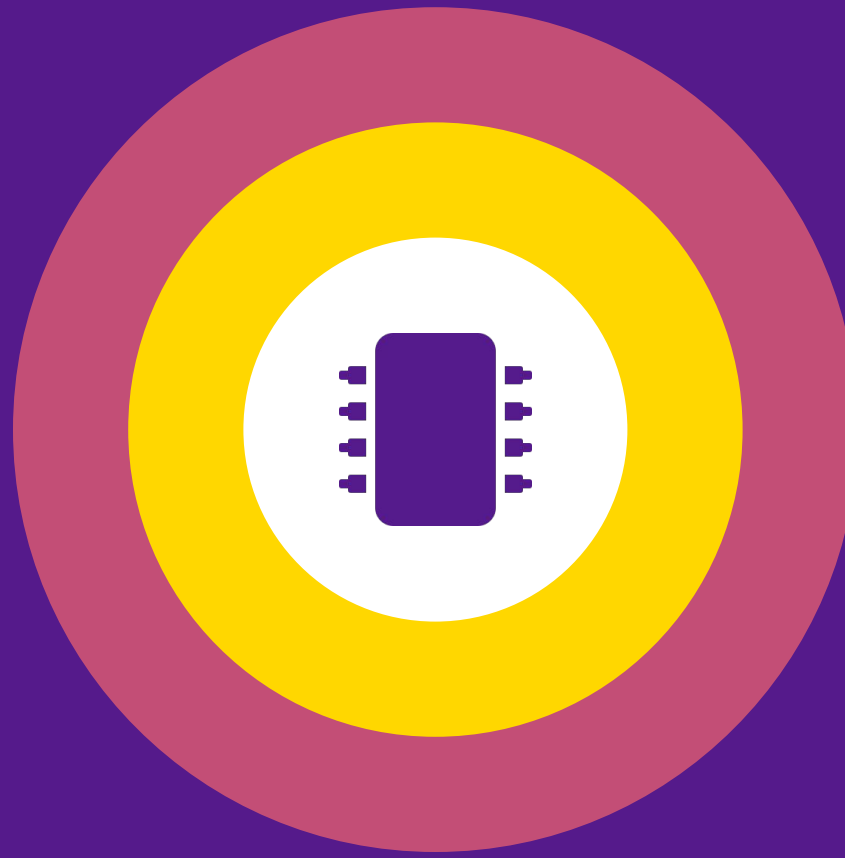
Thank you

Questions and feedback are welcome

Sense only when needed

Send only what matters

Save energy, keep accuracy



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